

ZEROPAY DIGITAL SERVICES LTD

United Kingdom Subsidiary of BETONIQ WEST LTD (RC 1496603, Nigeria)

AgriLink — Technical White Paper

Real-time farmer-to-buyer and cold-chain logistics marketplace.

Prepared by: Tom Ugbodaga — Shareholder & CEO

MSc Computer Systems Security (University of Gloucestershire — England, 2011)

LinkedIn: <https://www.linkedin.com/in/tom-deggs-ugbodaga-2306692bb>

September 2026

Prepared for grant, government, and institutional review

Executive Summary

AgriLink is a Zeropay Digital Services Ltd venture (UK subsidiary of BETONIQ WEST LTD) — Real-time farmer-to-buyer and cold-chain logistics marketplace.

Key statistic: 1.3 billion tonnes of food — roughly one third of everything the world produces — is lost or wasted every year, while 733 million people go hungry (FAO/UNEP data).

Problem Statement

Smallholder farmers in emerging markets routinely lose 30-40% of harvested crops before reaching a buyer, due to the absence of real-time market information, unreliable transport coordination, and multiple unnecessary broker layers that both delay sale and depress farmer income. AgriLink removes the information and logistics gap directly.

Technical Architecture

- Geolocation-ranked matching engine surfaces nearby listings to buyers and available trucks in real time.
- Lightweight mobile-first client designed for low-end Android devices and intermittent connectivity.
- Freshness-window scoring de-prioritizes listings approaching spoilage risk to encourage fast matching.
- In-app deal confirmation replaces informal broker negotiation with a timestamped digital record.

Core Data Model

Entity	Purpose
Farmer	Profile, location, active crop listings, verified transaction history.
CropListing	Crop type, quantity, harvest date, asking price, freshness window.
Buyer	Verified buyer/aggregator profile and purchase history.
LogisticsTruck	Real-time location, capacity, cold-chain capability, availability.
Transaction	Confirmed deal record: price, quantity, pickup time, settlement status.

Security, Privacy & Compliance

- Farmer and buyer personal data minimized to what is operationally necessary (name, phone, location).
- Transaction history immutable once confirmed, to support dispute resolution and lender credit assessments.
- Role-based access ensures buyers cannot see other buyers' bid data (sealed-bid style fairness).

Feasibility & Build Status

MVP data model and matching-engine logic specified. Frontend and API build proceeding on GitHub under Zeropay Digital Services Ltd; AI-assisted matching/ranking model hosted live on Hugging Face Spaces.

Market Opportunity

Global agri-supply-chain software market projected to exceed \$28B by 2030. Target beachhead: 2.5 million smallholder farmers across Nigeria's northern grain belt.

Grant & Funder Fit

Strong fit for agri-tech and food-security grant lines: IFAD Agri-Tech innovation funds, GSMA AgriTech Innovation Fund, EU Horizon Europe Food Systems calls, and national agriculture ministry digitization grants.

Funding Ask

\$150,000 seed grant to fund 12-month pilot across 3 Nigerian states (500 farmers, 50 buyers, 20 logistics partners).